Mediants and Collatz: Constructing Generalised Collatz Trajectories via Rational Approximation

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Abstract

We describe an application of the mediant tree algorithm of [1] to the construction of generalised Collatz trajectories. Given a generalised Collatz system (g, h) with odd step $x \mapsto gx + q$ and even step $x \mapsto x/h$ (where $h \nmid q$), a parity sequence realising a good rational approximant to $\theta = \log_h g$ is produced directly by the mediant tree as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$. We show how to convert such a mediant word into a concrete starting integer x_0 via a standard modular-arithmetic pipeline, and explain why the canonical greedy parity-sequence algorithm produces exactly the Sturmian word for slope $1/(1 + \log_h g)$. The parity–residue correspondence used in the pipeline is classical (Terras [2], Lagarias [3]); what is new is the *generative* role of the mediant tree as a transducer: it constructs the canonical balanced parity word and the rational approximant it realises in a single O(log N) integer-arithmetic pass, with no real arithmetic at runtime. We compare with Eliahou's continued-fraction bounding method [4], which provides lower bounds on Collatz cycle lengths via good approximants of $\log_2 3$ but gives no means of constructing a parity sequence realising any given approximant.

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1 Introduction

A *generalised Collatz sequence* is defined by a pair (g, h) with $g, h \geq 2$ and $\gcd(g, h) = 1$: an element x with $h \nmid x$ (an *odd* element) is mapped to $gx + q$ and an element with $h \mid x$ (an *even* element) is divided by h. The standard Collatz sequence corresponds to $g = 3$, $h = 2$, $q = 1$. The parity structure of a trajectory — the sequence of odd and even steps — is governed by $\theta = \log_h g$: a long-descending trajectory must have the ratio of even steps to odd steps approximate θ from above.

The question this paper answers. A non-trivial cycle in a Collatz system must have its parity sequence balanced so that the trajectory closes: following the cycle once gives $h^e x_0 = g^o x_0 + qK$, i.e. $(h^e - g^o)x_0 = qK$. The median element of such a cycle satisfies $\hat{x} = qK/(h^e - g^o)$, which is large precisely when $h^{e/o} \approx g$, i.e. when $e/o \approx \theta = \log_h g$. Since θ is irrational, $h^e \neq g^o$ always, but arbitrarily close rational approximants $e/o \approx \theta$ produce parity sequences where $\hat{x}$ is arbitrarily large. The mediant tree provides a canonical *generator* for such sequences: for any desired approximation quality, it produces an explicit e/o, the balanced parity word of length $e + o$ that realises it, and (via the pipeline of Section 4) a concrete starting integer x_0 whose trajectory follows that word exactly. This provides a systematic tool for exploring the structure of near-cycle parity sequences — the region of parity space where any non-trivial cycle, were it to exist, would have to live.

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The companion paper [1] develops the *mediant tree*. The present paper describes the Collatz application that originally motivated it. The main contributions are:

1. A pipeline (Section 4) for converting a mediant word into a concrete starting integer x_0 whose generalised Collatz trajectory follows the prescribed parity pattern for exactly $e + o$ steps.

2. A proof (Section 5) that the canonical greedy parity-sequence algorithm produces exactly the Sturmian word for slope $1/(1 + \log_h g)$ — the same word the mediant tree generates.

3. A discussion (Section 7) situating the work relative to Terras [2], Lagarias [3], and Eliahou [4], and explaining what the generative transducer adds over prior descriptive approaches.

Organisation. Section 2 recalls the key definitions and results from [1]. Section 3 introduces generalised Collatz sequences and parity sequences. Section 4 presents the step-by-step pipeline from mediant word to Collatz trajectory. Section 5 describes the greedy algorithm and its Sturmian interpretation. Section 6 discusses why general Collatz trajectories are not Sturmian. Section 7 compares with Eliahou's approach. Section 8 describes observations on

the carry polynomial's Laurent symmetry and future work directions: defect perturbation theory and palindromicity as an obstruction to limit cycles.

2 Preliminaries

We recall the definitions and results from [1] that are used in this paper. All notation is consistent with that reference.

A *ratio* is a pair $(e, o) \in \mathbb{N}^2$ with $\gcd(e, o) = 1$, written e/o. Given a real $\theta > 1$, a ratio $r = e/o$ is a *good estimate* of θ if $|e - \theta o| < 1$. Two good estimates $r_{2d} = e_{2d}/o_{2d}$ and $r_{2d+1} = e_{2d+1}/o_{2d+1}$ form a *bounded pair* (r_{2d}, r_{2d+1}) if they bracket θ and satisfy $\Delta(r_{2d}, r_{2d+1}) := e_{2d+1}o_{2d} - e_{2d}o_{2d+1} = \pm 1$.

Given a bounded pair (r_{2d}, r_{2d+1}) and a weight $w \geq 0$, the *weighted mediant* is

$$\text{Mediant}(r_{2d}, r_{2d+1}, w) = \frac{e_{2d} + w\, e_{2d+1}}{o_{2d} + w\, o_{2d+1}}.$$

The *optimal integer weight* w^+ is the unique integer such that the pair $\big(\text{Mediant}(r_{2d}, r_{2d+1}, w^-), \text{Mediant}(r_{2d}$ brackets θ, where $w^- = w^+ - 1$; it is given in closed form by Theorem 8.1 of [1], which also guarantees $w^+ \geq 1$.

A *leaf* estimate has denominator $o = 1$; its mediant word is $\mathsf{E}^e\mathsf{O}$ (Definition 9.1 of [1]). The children of a bounded pair at the optimal weight w^+ themselves form a bounded pair (Corollary 8.2 of [1]).

The *mediant word* $\pi(r)$ of a ratio r produced by the mediant tree is a string over $\{\mathsf{O}, \mathsf{E}\}$ of length $e + o$ encoding the sequence of weighted-mediant choices made to reach r from the initial bounded pair. It is a lower Christoffel word (Lemma 9.3 of [1]), ends in O (Theorem 12.1 of [1]), is OO-free (§12 of [1]), begins with EE for every non-leaf estimate (Theorem 12.4 of [1]), and is identical to the Sturmian word for slope $1/(1 + \theta)$ at all positions except possibly the last (Theorem 12.6 of [1]).

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3 Generalised Collatz Sequences and Parity Sequences

A *generalised Collatz sequence* uses an odd step $x \mapsto gx + q$ and an even step $x \mapsto x/h$, where $g, h \in \mathbb{N}$ with $g, h \geq 2$ and $\gcd(g, h) = 1$, and $q \in \mathbb{Z}$ with $q \neq 0$ and $h \nmid q$. The parity sequence structure depends only on g and h through $\theta = \log_h g$; the value of q affects only the concrete starting residue x_0 (Step 4 of the pipeline in Section 4). A *parity sequence* is a string over $\{\mathsf{O}, \mathsf{E}\}$ recording the sequence of odd and even steps along a trajectory; O denotes an odd step and E an even step.

For a trajectory to descend on average, the ratio of even steps to odd steps must exceed $\log_h g$: if a segment of length $e + o$ has e even steps and o odd steps, the net effect is multiplication by g^o/h^e, which is less than 1 iff $e/o > \log_h g$. The boundary case $e = \lceil o \log_h g \rceil$, o fixed, gives a good estimate e/o of $\log_h g$ (when $\gcd(e, o) = 1$), and corresponds to the upper-bound condition

$$h^{e-1} < g^o < h^e.$$

A parity sequence realising a good estimate e/o must therefore have exactly o copies of O and e copies of E. The mediant word generator produces OO-free words by construction, as does the Sturmian word for slope $\alpha < 1/2$ (equivalently $\theta > 1$). For $h = 2$, OO-freeness is also a natural consequence of the deterministic Collatz map: each odd step produces an even result, forcing at least one E before the next O. Because every mediant word is OO-free, it is realisable as a deterministic $h = 2$ Collatz trajectory [5].[1]

4 From Mediant Word to Collatz Trajectory: the Pipeline

The mediant tree applies to any real $\theta > 1$, not only to logarithms. The following pipeline converts a mediant word into a concrete Collatz starting integer for any (g, h) with $\theta = \log_h g$.

Step 1: Choose a real θ and an initial bounded pair. Select any real $\theta > 1$ and a Farey-adjacent bounded pair (r_0, r_1) with one estimate on each side of θ, with r_1 the better estimate (i.e. $|\varepsilon_{r_1}| < |\varepsilon_{r_0}|$). For the standard Collatz system take $\theta = \log_2 3$; for the generalised system (g, h) take $\theta = \log_h g$. The construction applies equally to any irrational θ (e.g. π, $\sqrt{2}$), in which case it produces rational approximants without a Collatz interpretation.

Step 2: Run the mediant tree. Apply the optimal weight formula (Theorem 8.1 of [1]) to produce a sequence of mediant words $\pi(r_{2d+1})$ of increasing length. At depth d, the word $\pi(r_{2d+1})$ has o_{2d+1} odd steps and e_{2d+1} even steps, and e_{2d+1}/o_{2d+1} is a good estimate of θ satisfying $h^{e-1} < g^o < h^e$ (upper bound) or $h^e < g^o < h^{e+1}$ (lower bound).[2]

Step 3: Derive the path identity. For a chosen mediant word $\sigma = \pi(r_{2d+1})$, the function `path_identity(`σ`)` constructs the *basic path identity*

$$h^e\, x_n = g^o\, x_0 + q\, K(g, h),$$

where x_0 is the starting value, x_n is the value after $n = e + o$ steps following σ, and $K(g, h)$ is the *carry polynomial* — a polynomial in g and h whose coefficients record the cumulative effect of the additive terms along the trajectory, with q scaling the carry to the chosen additive constant. This identity holds for every $t \in \mathbb{N}_0$: for each non-negative integer t, the pair $(x_0 + h^{e+1}t,\ x_n + g^o h\, t)$ is another valid (start, end) pair following σ, as seen from the equivalent *shifted path identity*

$$h^e\big(x_n + g^o h\, t\big) = g^o\big(x_0 + h^{e+1}t\big) + q\, K(g, h),$$

which makes explicit that the family of valid (x_0, x_n) pairs forms an arithmetic progression with step $(h^{e+1}, g^o h)$.

Step 4: Instantiate to concrete (g, h) and solve for x_0. The function `path_identity_params(`σ`, `g`, `h`)` substitutes the concrete multiplier and base, computes K, and solves

$$x_0 \;\equiv\; -q\, K \cdot (g^o)^{-1} \pmod{h^{e+1}}$$

[1] Non-deterministic $h = 2$ Collatz maps — where the odd/even instruction is supplied externally rather than derived from the state — do permit adjacent OO symbols; see [7].

[2] These inequalities follow from the good-estimate condition $|e - \theta o| < 1$ with $\theta = \log_h g$: an upper estimate has $e - 1 < \theta o < e$, giving $h^{e-1} < h^{\theta o} = g^o < h^e$; a lower estimate has $e < \theta o < e + 1$, giving $h^e < g^o < h^{e+1}$. The general statement was verified by machine-checked proof in Lean 4 by an AI assistant; see `bounded_pair_power_bounds` in the companion Lean development.

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for a starting residue $x_0 \in [0, h^{e+1})$. x_0 is an element whose parity sequence follows σ for exactly $o + e$ steps and then continues with an E (an *even-continuing* residue). In total, $h - 1$ residues mod h^{e+1} follow σ; the formula selects the unique even-continuing one, and the other $h - 2$ continue with an O.[3]

Step 5: Read off the approximation data. The output tuple $(o,\ e,\ e/o,\ x_0)$ encodes:

- e/o — a rational approximant to θ, with $e/(o+e)$ and $o/(o+e)$ the even and odd step densities;
- x_0 — the even-continuing starting residue (mod h^{e+1}) whose $gx+q$ trajectory realises the mediant word σ for exactly $o + e$ steps.

Setting $t = 0$ gives $x_n = (g^o x_0 + qK)/h^e$, which is also determined by x_0. As $d \to \infty$ the ratio e_{2d+1}/o_{2d+1} converges to θ, with convergence rate governed by the continued-fraction expansion of θ.

Remark. Evaluating the Collatz iteration from x_0 is *not* an efficient way to generate the parity sequence: it requires arithmetic with integers of size $O(h^e)$, which grows exponentially in the word length. The parity sequence itself is produced directly and efficiently by the mediant tree, with no large integer arithmetic. The value of x_0 lies in establishing that a concrete Collatz trajectory realising the sequence exists — not in using it to regenerate the sequence.

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Remark (residue constraint). To the extent that a $3x + 1$ trajectory matches a factor of the Sturmian word for $\theta = \log_2 3$, every odd element of the trajectory is congruent to 1 or 3 modulo 8. This is a consequence of the Sturmian alphabet being $\{\mathsf{EO}, \mathsf{EEO}\}$: an odd step from a residue $r \equiv 1 \pmod 8$ produces $v_2(3r+1) = 2$ even steps (an EEO block), while $r \equiv 3 \pmod 8$ produces $v_2(3r + 1) = 1$ even step (an EO block); residue 5 (mod 8) forces the parity sequence $\mathsf{OEEE}\cdots$, and residue 7 (mod 8) forces the parity sequence OEOEO; neither pattern occurs in the Sturmian word for $\log_2 3$, so neither residue can appear in a trajectory matching it. A full account is given in [6].

5 The Greedy Parity Sequence Algorithm

For any generalised Collatz system (g, h) with $g, h \geq 2$ and $\gcd(g, h) = 1$, there is a simple greedy algorithm that generates the idealised parity sequence using only integer arithmetic — no floating-point evaluation of $\log_h g$ is required. Let $short = \lfloor \log_h g \rfloor$ and $long = short+1$ (the two possible E-run lengths, since $\log_h g \in (short, long)$).

```
def collatz_log_h_of_g(g=3, h=2):
    e=o=0
```

[3] That $h^{e+1} \mid g^o x_0 + qK$ is exactly the condition for x_0 to follow σ and continue with E (when $h \nmid q$) was verified by machine-checked proof in Lean 4 by an AI assistant; see `step4_even_continuing_iff` in the companion Lean development. The counts ($h - 1$ total followers, 1 even-continuing, $h - 2$ odd-continuing) are proved in `step4_count_followers`, `step4_count_even_continuing`, and `step4_count_odd_continuing`. For $h = 2$ uniqueness follows since $h - 1 = 1$.

```
short=0
while h**(short+1) <= g:
    short+=1
long=short+1
while True:
    if g**(o+1) < h**(e+short):
        yield from 'E' * short
        yield 'O'
        e+=short
        o+=1
    else:
        yield from 'E' * long
        yield 'O'
        e+=long
        o+=1
```

The condition $g^{o+1} < h^{e+short}$ is a pure integer comparison equivalent to $(o+1)\log_h g < e + short$, i.e. appending a *short* block $\mathsf{E}^{short}\mathsf{O}$ is sufficient to keep the ratio e/o above $\log_h g$; otherwise a *long* block $\mathsf{E}^{long}\mathsf{O}$ is needed. This is exactly the Beatty/Sturmian greedy rule: at each O step the algorithm selects the shorter run when the fractional part of $(o+1)\log_h g$ is already covered by the accumulated e, and the longer run otherwise.

The resulting sequence is the Sturmian word for slope $\alpha = 1/(1+\log_h g)$, characterised by $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ for all $n \geq 0$.

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Proof that the greedy output equals the Sturmian word. Write $\theta = \log_h g$ (irrational: if $\log_h g = a/b$ then $g^b = h^a$, impossible for coprime $g, h \geq 2$) and $s = \lfloor \theta \rfloor$. Let e_k denote the total number of E-steps output after k complete O-steps. We claim that $e_k = \lceil k\theta \rceil$ for all $k \geq 0$.

Base case. $e_0 = 0 = \lceil 0 \rceil$. ✓

Inductive step. Assume $e_k = \lceil k\theta \rceil$. Since $\lceil k\theta \rceil - 1 < k\theta \leq \lceil k\theta \rceil$ and $s \leq \theta < s+1$, we have

$$\lceil k\theta \rceil + s - 1 < (k+1)\theta < \lceil k\theta \rceil + s + 1.$$

The greedy appends a *short* block (s copies of E then O) iff $g^{k+1} < h^{e_k+s}$, equivalently $(k+1)\theta < e_k + s = \lceil k\theta \rceil + s$. In that case $\lceil (k+1)\theta \rceil = \lceil k\theta \rceil + s$, so $e_{k+1} = e_k + s = \lceil (k+1)\theta \rceil$. Otherwise $(k+1)\theta > \lceil k\theta \rceil + s$ (strictly, since θ is irrational), so $\lceil (k+1)\theta \rceil = \lceil k\theta \rceil + s + 1$, and $e_{k+1} = e_k + s + 1 = \lceil (k+1)\theta \rceil$. Either way $e_{k+1} = \lceil (k+1)\theta \rceil$, completing the induction.

Now the k-th O (1-indexed) is emitted at position $e_k + k - 1$ (0-indexed) in the output. With $e_k = \lceil k\theta \rceil$ this is position $\lceil k\theta \rceil + k - 1 = \lfloor k(1+\theta) \rfloor$. The Sturmian word has $s_n = \mathsf{O}$ at position n iff $n = \lfloor k(1+\theta) \rfloor$ for some $k \geq 1$, which matches exactly. Since a binary word over $\{\mathsf{O}, \mathsf{E}\}$ is uniquely determined by the index positions of its O's, the greedy output and the Sturmian word are identical. □

For the standard case $g = 3$, $h = 2$ the first 60 symbols are:

EEOEEOEOEEOEOEEOEEOEOEEOEOEEOEEOEOEEOEOEEOEOEEOEEOEOEEOEOEEO,

with O-density converging to $1/(1+\log_2 3) = \log_6 2 \approx 0.387$ — the *ideally balanced* parity pattern, descending as slowly as theory permits. For $g = 5$, $h = 2$ (*short* = 2, *long* = 3, since $\log_2 5 \approx 2.322$) the first 60 symbols are:

EEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEOEEEOEEOEEO,

with O-density converging to $1/(1+\log_2 5) \approx 0.301$. The $5x+1$ system has three known cycles; the mediant word describes the idealised parity pattern of a trajectory that descends at the asymptotically optimal rate for that system.

6 General Collatz Sequences are not Sturmian

The Sturmian word for slope $\alpha = 1/(1+\theta)$ consists of exactly two block types: $\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil\theta\rceil}\mathsf{O}$, interleaved in the Sturmian pattern. These are precisely the leaf words of the mediant tree for θ. A Sturmian word can never contain a block of any other length.

A general Collatz trajectory is far less constrained: an actual orbit can produce E-runs of any length, interleaved freely. The mediant word (equivalently, the Sturmian word) is the *ideally balanced* special case.

Moreover, even among trajectories that approximate $\log_h g$ closely, the Sturmian alphabet $\{\mathsf{E}^{\lfloor\theta\rfloor}\mathsf{O}, \mathsf{E}^{\lceil\theta\rceil}\mathsf{O}\}$ is not the only primitive alphabet that can arise. For example, the three known cycles of the $5x+1$ system have alphabets $\{\mathsf{EO}, \mathsf{EEEEO}\}$, $\{\mathsf{EO}, \mathsf{EEEEEO}\}$, and $\{\mathsf{EO}, \mathsf{EEEO}\}$ — each containing the block EO of length 2, which does not appear in the Sturmian alphabet $\{\mathsf{EEO}, \mathsf{EEEO}\}$ at all. The mediant tree generates parity sequences over the Sturmian alphabet only; it produces the optimal approximation to θ but does not exhaust all primitive alphabets whose e/o ratio approximates θ.

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7 Relationship to Eliahou's Work

Classical context. The connection between parity sequences and rational approximation of $\log_h g$ is classical. Terras [2] introduced the parity sequence as a tool for analysing Collatz orbits and observed that a descending trajectory requires the ratio of even to odd steps to exceed $\log_2 3$. Lagarias [3] gave a systematic account of the parity–residue correspondence, relating the number of odd and even steps in a segment to the residue class of the starting integer modulo a power of h. Eliahou [4] sharpened this to explicit lower bounds on nontrivial cycle lengths, using best rational approximants of $\log_2 3$ obtained from the continued-fraction expansion.

These works are *descriptive*: they characterise what parity sequences must look like and what constraints a cycle must satisfy, but they do not provide a means of constructing a parity sequence achieving a given approximation quality.

The generative transducer. The contribution of the present work is a *generative* one. The mediant tree of [1] is a transducer: given a target $\theta = \log_h g$, it simultaneously constructs a rational approximant e/o to θ and the canonical balanced parity word realising it, in a single pass using only integer arithmetic. No real arithmetic is required at runtime; the

approximation quality is controlled entirely by the depth to which the tree is expanded. The mediant word is not just any parity word with the right ratio of E's to O's: it is the *canonical* one, identical to the Sturmian word for slope $1/(1+\theta)$ at all interior positions (Theorem 12.6 of [1]), and palindromic in its interior (Theorem 12.5 of [1]). These structural properties make it the unique *lower Christoffel* representative of its approximation class.

Comparison with Eliahou. Our algorithm constructs approximants directly from bounded pairs via the optimal weight formula (Theorem 8.1 of [1]), without continued-fraction computation; the continued-fraction connection emerges as a consequence of the Sturmian identity (Section 10 of [1]). Eliahou's framework gives lower bounds on cycle lengths but provides no construction of a parity sequence realising any given approximant.

The present work generalises in both directions: the mediant tree applies to any real θ (not just $\log_2 3$), and the pipeline of Section 4 produces a concrete trajectory for any generalised Collatz system (g, h, q) — with the system parameters chosen freely to match the approximation target.

8 Observations and Future Work

8.1 Laurent symmetry of the carry polynomial

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The palindrome theorem for mediant words (Theorem 12.5 of [1]) has a direct algebraic consequence for the carry polynomial $K(g, h)$.

Recall that the path identity for a word σ of length $e + o$ is

$$h^e\, x_n = g^o\, x_0 + q\, K(g, h),$$

where $K(g,h) = \sum_{j:\,\mathsf{O}_j} g^{a_j} h^{b_j}$ records the cumulative additive contributions, with one term per O in σ. The exponent pair (a_j, b_j) at O-position j counts: a_j = number of O's *after* position j; b_j = number of E's *before* position j.

Every mediant word has the form $\sigma = \mathsf{E} \cdot P \cdot \mathsf{O}$ where P is a palindrome (Theorem 12.5 of [1]). The leading E contributes no O, but shifts all b_j up by 1; the trailing O contributes the term $g^0 h^e = h^e$ (all e even steps lie before it). Separating this unpaired term, write

$$K(g, h) = h^e + C(g, h),$$

where $C = gh \cdot K_P(g, h)$ absorbs the contributions from the interior O's. The palindrome symmetry of P forces the exponent pairs of C to come in matched pairs: if (a, b) appears in C, so does $(o - a,\ e + 1 - b)$. Equivalently, C satisfies the self-reciprocal identity

$$C(g, h) = g^o h^{e+1}\, C(g^{-1}, h^{-1}),$$

so centering by $g^{o/2} h^{(e+1)/2}$ yields a Laurent polynomial in $g^{1/2}, h^{1/2}$ invariant under simultaneous inversion. Lean: `laurent_symmetry`, `TreeWord.laurent_symmetry`.

This structural property — that the carry polynomial of any mediant word is a symmetric Laurent polynomial after central rescaling — implies that the cumulative additive

contributions *accumulate from the start to the midpoint* in a pattern that is the exact mirror image of the *descent from the midpoint to the end*. The palindrome in the word domain acts as a conservation law in the algebraic domain, constraining $K(g, h)$ far more tightly than an arbitrary parity word of the same length.

A complete treatment (precise statement, proof, and implications for the structure of K for different depths of the mediant tree) is left as future work.

8.2 Defect perturbation theory

The mediant tree generates the *optimal* parity sequence for a given approximation quality: a word that is Sturmian, palindromic, and OO-free. Any actual Collatz trajectory that is not Sturmian can be viewed as a perturbation of this optimal backbone.

Specifically, a *defect* is any position where the trajectory uses a block of length $\ell \notin \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$ (i.e. a block length not in the Sturmian alphabet for θ). Each defect at word position k perturbs the carry polynomial $K(g, h)$ globally: replacing a Sturmian block $\mathsf{E}^\ell\mathsf{O}$ with an out-of-alphabet block $\mathsf{E}^{\ell'}\mathsf{O}$ (where $\ell' \notin \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$) changes e by $\ell' - \ell$, which shifts b_j by $\ell' - \ell$ for every O at position $j \geq k$, and changes the modulus h^{e+1}. The total perturbation to K is therefore a sum over all subsequent O positions, not a local change at k alone. Lean: `carry_insert_E`.

Because the starting residue is determined by $x_0 \equiv -qK(g^o)^{-1} \pmod{h^{e+1}}$, every defect causes a predictable arithmetic shift in the required x_0. This gives the mediant tree the role of a *reference frame*: the distance of any Collatz trajectory from the optimal backbone can be measured by the cumulative drift in x_0 induced by its defects.

This perturbation-theory perspective suggests a systematic approach to the structure of non-Sturmian Collatz trajectories. In particular, the known non-trivial cycles of the $5x + 1$ system (Section 6) each have a specific defect structure relative to the Sturmian alphabet $\{\mathsf{EEO}, \mathsf{EEEO}\}$; computing the drift each defect induces, and the corresponding shift in x_0, would quantify how far each cycle sits from the optimal backbone. A full development of this defect perturbation theory — including a formula for the drift as a function of defect type and position, and a classification of which drift patterns are compatible with the modular closure constraint for a cycle — is left as future work.

8.3 Palindromicity as an obstruction to limit cycles

The palindrome invariant (Theorem 12.5 of [1]) characterises the *family* of optimal descent paths generated by the mediant tree. While a pure Sturmian word is an infinite, aperiodic object, finite factors of mediant words can appear as parity sequences of cycles in related systems (such as the $3x - 1$ cycle, whose parity period is a cyclic rotation of a mediant word factor). Nonetheless, the palindromic structure of mediant words provides a powerful algebraic reference frame for analysing candidate cycles.

Specifically, the palindrome invariant acts as a conservation law on the carry polynomial $K(g, h)$, enforcing a symmetric Laurent structure (Section 8.1). Any actual cycle in a generalised Collatz system must deviate from this ideal mediant backbone by incorporating "defects"—local variations in block lengths outside the pure Sturmian alphabet $\{\lfloor\theta\rfloor, \lceil\theta\rceil\}$.

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The mediant framework suggests a two-part perspective on why non-trivial cycles are so rigidly constrained:

1. **Optimality versus exact closure.** The mediant tree constructs the maximally balanced parity sequences that asymptotically approach optimal descent for slope $\alpha = 1/(1+\theta)$. A true cycle requires an exact integer closure condition: from the path identity $h^e x_n = g^o x_0 + qK(g,h)$, setting $x_n = x_0$ yields the exact integer equation

$$(h^e - g^o)x_0 = qK(g,h).$$

   Because $\theta = \log_h g$ is irrational, $h^e \neq g^o$ for any finite word length, meaning any exact cycle must inhabit a precise arithmetic balance between the multiplicative drift $(h^e - g^o)$ and the carry polynomial K.

2. **Defect drift and algebraic obstruction.** Because actual cycle words deviate from the pristine palindromic mediant backbone, their carry polynomials experience algebraic "drift" away from the symmetric Laurent structure. Each defect perturbs the coefficients of $K(g,h)$ in a measurable way. For a candidate cycle to satisfy the exact integer equation $(h^e - g^o)x_0 = qK$, the cumulative drift induced by these defects must conspire exactly with the multiplicative mismatch. Whether the algebraic constraints imposed by the palindromic conservation law forbid such defect configurations from ever satisfying the exact cycle equation remains an open question.

Rather than viewing palindromicity as a rigid combinatorial barrier that forbids finite-factor cycles outright, the mediant framework provides a quantitative baseline: it measures exactly how far any candidate cycle deviates from the symmetric, asymptotically optimal backbone, casting cycle existence as a question of whether defect-induced algebraic drift can satisfy exact integer closure.

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References

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